

BRIGHTTV VIEWERSHIP ANALYTICS

Audience Behaviour, Content Consumption & Customer Growth Opportunities

Project Description

This Data Analytics case study analyses BrightTV viewing-session data to understand audience behaviour, content consumption patterns, viewing timing and geographic distribution. The analysis translates observed viewing patterns into Customer Value Management (CVM) opportunities for engagement and user growth.

1. Project Background

BrightTV wants to grow its subscription base. The CVM team therefore needs evidence about who watches, what they watch, when they watch and where users are located. This project uses the available viewing-session data to identify actionable patterns and opportunities.

2. Business Problem

The central business problem is to understand viewer consumption behaviour and use that evidence to support content, engagement and customer-growth decisions.

3. Project Objectives

1. Identify the main user and usage trends.
2. Identify factors associated with consumption.
3. Recommend content strategies for low-consumption days.
4. Recommend initiatives that could further grow the BrightTV user base.

4. Dataset Overview

The project uses BrightTV viewing-session data containing user, demographic, channel and date/time information. Each viewing session is represented as one record. The analysis includes User ID, Age, Gender, Province, Channel and viewing date/time attributes.

5. Methodology

The workflow included data inspection and preparation, handling of unknown values, age-group creation, user/session aggregation, channel analysis, day-of-week analysis and hourly analysis. UTC timestamps were converted to South African Standard Time (SAST) so that viewing patterns could be interpreted using local time.

Databricks and SQL were used for the core data processing and analytical queries. The findings were then communicated through visual analysis, an interactive dashboard and a final presentation.

6. Key Findings

Metric	Verified result
Unique users	4,386
Total sessions	9,995
Sessions per user	2.28
Largest age group	25–34 (40.56% of users)
25–44 combined	67.53% of users

Metric	Verified result
Male users	84.88%
Female users	11.13%
Highest viewing day	Friday – 1,673 sessions
Lowest viewing day	Monday – 956 sessions
Peak viewing hour	17:00 – 648 sessions
Largest observed province	Gauteng – 1,607 users
Top channel	Supersport Live Events – 1,637 sessions

7. Consumption Analysis

Consumption was examined across age, gender, province, channel, day and hour, as well as age-by-gender and channel-by-gender combinations. The analysis identifies observed associations and patterns; it does not establish causal relationships.

8. Low-Consumption Day Opportunity

Monday was the lowest-consumption day with 956 sessions and 769 unique users. Monday's strongest channels included Channel O (136 sessions), Trace TV (115), Supersport Live Events (98), Cartoon Network (94) and Africa Magic (92). These patterns provide a basis for testing Monday-specific programming and engagement strategies.

9. Business Recommendations

- Prioritise the 25–44 audience for targeted engagement and retention initiatives.
- Test strategies aimed at improving female audience acquisition and engagement.
- Explore referral, advocacy and social-sharing initiatives among highly engaged younger audiences.
- Explore regional targeting using observed audience concentration, particularly Gauteng.
- Strategically promote strong sports and event content.
- Test Monday-specific music, entertainment and selected sports/event programming, supported by targeted reminders or promotions.

10. Dashboard & Presentation

An interactive BrightTV dashboard was developed to communicate the findings through Overview, Audience, Viewing Behaviour, Channel Performance, Demographic Analysis, Detailed Analysis, Key Insights and Data Methodology views. The project also includes a final presentation designed for an approximately 20-minute business/academic presentation.

11. Data Quality & Limitations

The dataset contains unknown or missing demographic values and channel naming inconsistencies, including “SawSee” versus “Sawsee” and “Supersport Live Events” versus “SuperSport Live Events”. These issues were identified and flagged. Provincial audience size should not be interpreted as market penetration because population or subscriber denominator data was not available. Smaller segments should also be interpreted cautiously. Recommendations should be validated through controlled testing and measurement.

12. Conclusion

The analysis demonstrates how BrightTV viewing-session data can be converted into practical CVM insights. The strongest observed patterns include a core 25–44 audience, high male representation, concentrated viewing in Gauteng, Friday as the strongest viewing day, Monday as the weakest, a 17:00 peak and strong performance from sports/event and entertainment channels. These findings provide a foundation for targeted engagement, content experimentation and customer-growth initiatives.